Course Code	CS 448/648
Title of the Course	Software Verification and Certification
Course Category	Department Elective
Credit Structure	L-T-P–Credits 2-1-0-3
Department	Computer Science and Engineering
Pre-requisite	Knowledge of Automata Theory and Logic, Computer Programming
Objectives	1. To understand the significance of correctness, safety, and security in open-source software. 2. To learn techniques of formal verification and static code analysis.
Outcomes	1. Apply formal verification methods to ensure software safety, liveness, and correctness. 2. Embed correctness checks throughout the software lifecycle for certified and responsible development.
Course Syllabus	• **Foundations of Software Correctness:** Fundamentals of software correctness-safety-liveness, formal software verification, verification techniques: deductive vs algorithmic and hybrid approaches • **Formal Verification Techniques:** Model checking and Theorem proving, Program analysis: Static vs dynamic, Case studies, synergistic verification and validation • **Verification- Languages, Tools, and Standards:** property specification languages, system modelling languages, and tools, domain specific Coding standards and implications • **Collaborative Verification and Licensing Considerations:** openness vs verifiability, Industrial case studies, Application in responsibility specification and analysis • **Correctness-Driven Software Lifecycle:** Integration of correctness in the software development lifecycle, correctness-certification of open-source projects
Suggested Books	Textbooks: 1. Edmund M. Clarke Jr., Orna Grumberg, Daniel Kroening, Doron Peled and Helmut Veith, "**Model Checking",** the MIT Press, 2018, ISBN: 9780262038836 2. Xavier Rival, Kwangkeun Yi, "**Introduction to Static Analysis: An Abstract Interpretation Perspective",** The MIT Press, 2020, ISBN:9780262043410 Reference books: 3. Steven Weber, "**The Success of Open Source**", Harvard University Press, USA, 2005, ISBN: 9780674018587